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Experiment Number : 3

Experiment Title : Simulation of Signal Transmission through Wired Media

Date of Submission : 02/04/2026

Experiment No. 03

Title: Simulation of Signal Transmission through Wired Media

Aim

To study and simulate the characteristics of signal transmission through a lossy wired medium (Coaxial Cable) using LTspice and observe the effects of attenuation and propagation delay.

Software Used

LTspice 26.0.1 (High-performance SPICE simulation software).

Theory

Wired Media Characteristics: In telecommunication, physical cables like Coaxial or Twisted Pair are not ideal conductors. They possess electrical properties distributed along their length: Resistance (R), Inductance (L), Capacitance (C), and Conductance (G).

The LTRA Model: The Lossy TRAnsmission line (LTRA) model in LTspice simulates real-world behavior.

Attenuation: The signal strength decreases over distance due to Resistance (R).

Distortion: The signal shape changes (rounding of edges) due to Capacitance (C) and Inductance (L).

Propagation Delay: It takes a finite amount of time for the signal to travel from the source to the destination.

Simulation Procedure

Step 1: Component Selection and Placement

Open LTspice and create a New Schematic.

- *Voltage Source:* Select voltage, and place it. Right-click it, select Advanced, then PULSE. Set: Vinitial=0, Von=5, Tdelay=0, Trise=1n, Tfall=1n, Ton=0.5u, Tperiod=1u.

- *Transmission Line:* Press 'P' search for the component 'ltline'. Select it and place it on the schematic. It will appear with the designator O1 and LTRA.
- *Load Resistor:* Press 'R', place a resistor at the output. Right click on the resistor symbol and set resistance value to 50 ohms (to match the characteristic impedance of co-axial cable).

Step 2: Defining the Cable Model

- Right-click the ltline (O1) component. In the Value field, replace LTRA by typing 'MyCo-axCable'.
- Click the '.t' button (SPICE Directive) on the top toolbar.
- In the dialog box, paste the following command exactly: .model MyCo-axCable LTRA(R=0.1 L=250n C=100p G=0 len=100)
- Place this text anywhere on your schematic area. This defines MyCo-axCable as a 100-meter lossy line.

Step 3: Wiring and Grounding

- Press 'G' to place grounds. You must ground the bottom terminal of the Voltage Source, the two bottom terminals of the ltline (O1), and the Load Resistor.
- Press 'W' to wire the components. Connect all points properly.
- Use 't' to label the input wire as IN and the output wire as OUT.

Step 4: Setting up the Simulation

- Click configure analysis (.ac) from the toolbar or type 'A' to configure transient analysis.
- In the Transient tab, set the Stop Time to 5u.
- Place the .tran 5u command anywhere on the schematic.
- Click the Run (Running Man) icon.

Observation and Analysis

Students must perform the simulation for the following three cases by right-clicking the .model directive and changing the len value each time.

Case 1: Cable Length = 100 meters

- Task: Run simulation with len=100.
- Screenshot: Capture the graph showing both IN and OUT waveforms. Paste the screenshot image of waveforms here.
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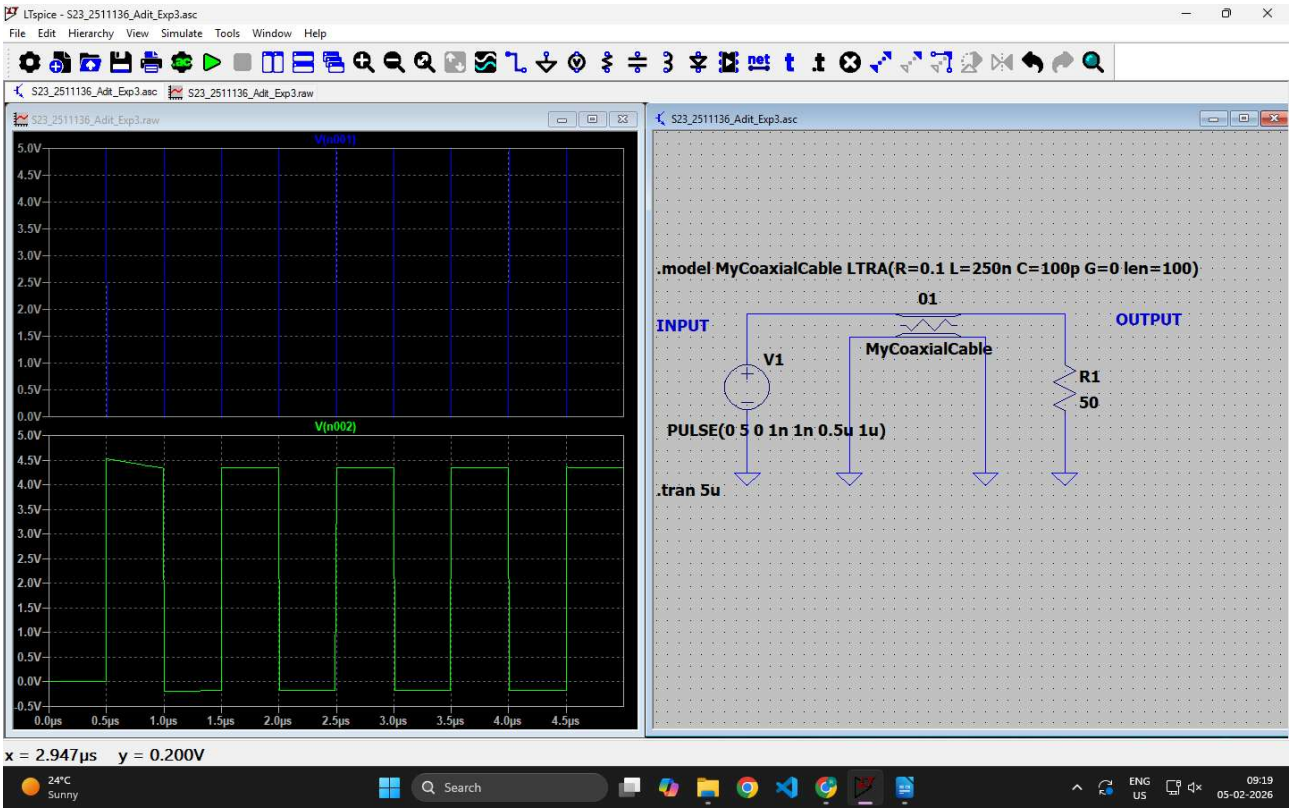
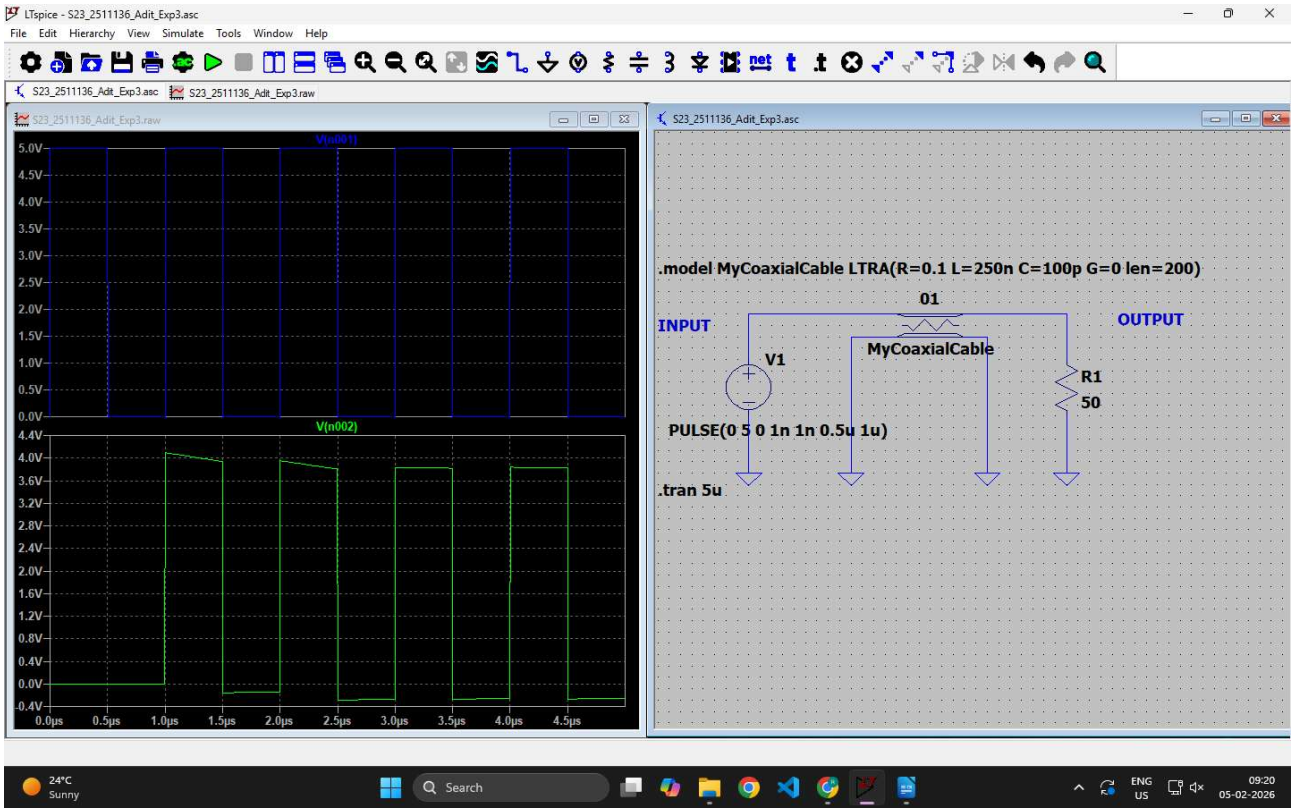


- Data: Record the peak output voltage and the time delay.

Case 2: Cable Length = 200 meters

- Task: Update directive to len=200 and re-run. Change the SPICE Directive to define cable length of 200 meters
  .model MyCo-axCable LTRA(R=0.1 L=250n C=100p G=0 ***len=200****)*

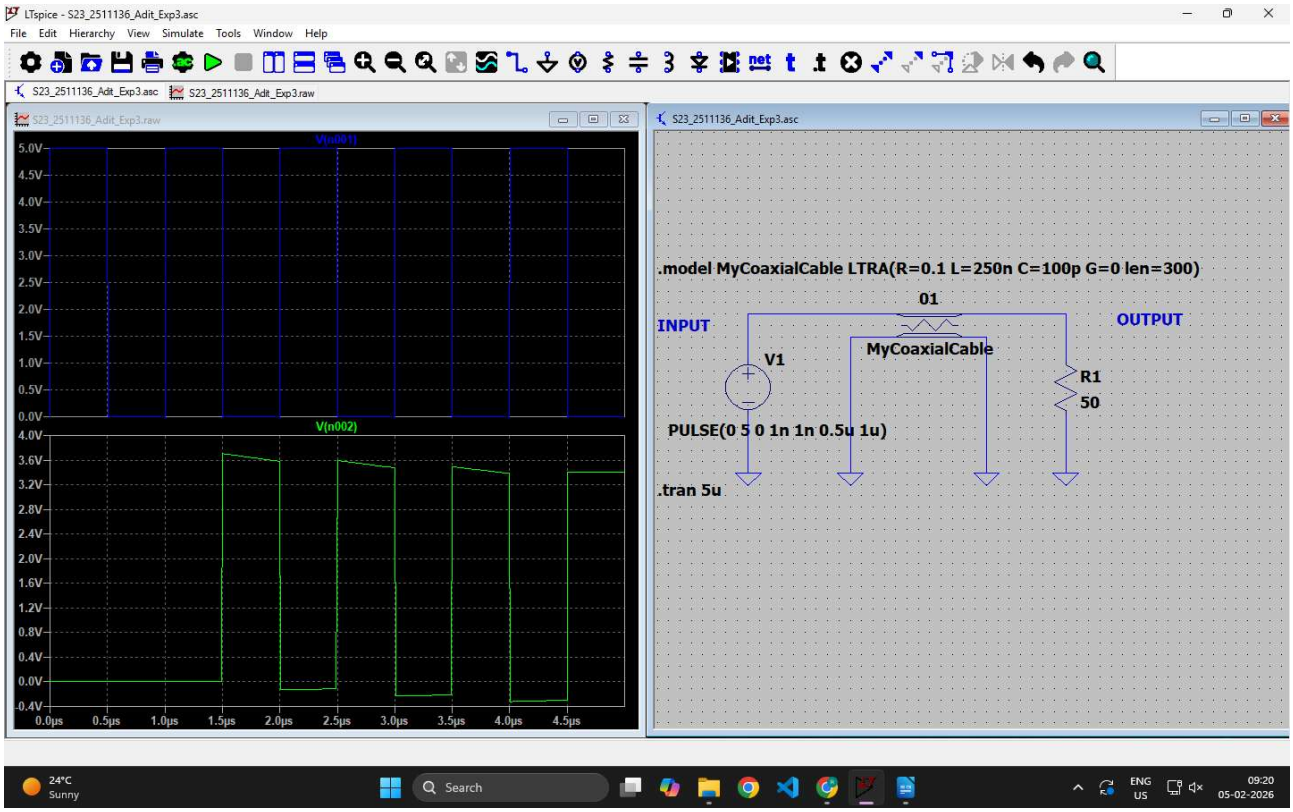
- Screenshot: Capture the graph showing both IN and OUT waveforms. Paste the screenshot image of waveforms here.



- Data: Record the peak output voltage and observe the increased delay.

Case 3: Cable Length = 300 meters

- Task: Update directive to len=300 and re-run. Change the SPICE Directive to define cable length of 300 meters
  .model MyCo-axCable LTRA(R=0.1 L=250n C=100p G=0 ***len=300****)*
- Screenshot: Capture the graph showing both IN and OUT waveforms. Paste the screenshot image of waveforms here.
- Data: Record the peak output voltage and observe the increased dealy.



Observation Table

Case	Cable Length	Input Voltage	Output Voltage	Propagation Delay
1	100 m	5V	4.4V	0.5us
2	200 m	5V	3.9V	1us
3	300 m	5V	3.5V	1.5us

Conclusion

The simulation demonstrates that wired media is not ideal. As the cable length increases, attenuation and propagation delay become more prominent. For the cable length of 100 m, 200 m and 300 m the attenuation observed are 12%, 22%, and 30% respectively whereas propagation delay observed was 0.5μs, 1μs and 1.5μs respectively.